

Section : Chemistry

Q.1 Pick the reactions which can be used to distinguish between n-propyl alcohol and isopropyl alcohol.

- (A) Haloform reaction
- (B) Dehydration
- (C) Dehydrogenation with Cu/573 K
- (D) Oxidation with CrO_3

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (C) and (D) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 The IUPAC name of the complex $\text{K}_3[\text{Fe}(\text{CN})_5\text{NO}]$ is

- 1. Potassium pentacyanonitrosyliron(II)
- 2. Potassium pentacyanonitrosylferrate(II)
- 3. Potassium nitrosylpentacyanoferrate(II)
- 4. Potassium pentacyanonitrosylferrate(III)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Match the complexes given in List-I with their properties given in List-II

- | List-I
Complex | List-II
Property |
|-------------------------------------|------------------------|
| (A) $[\text{FeF}_6]^{3-}$ | (I) d^6 , high spin |
| (B) $[\text{CoF}_6]^{3-}$ | (II) d^5 , high spin |
| (C) $[\text{Mn}(\text{CN})_6]^{3-}$ | (III) d^6 , low spin |
| (D) $[\text{Co}(\text{CN})_6]^{3-}$ | (IV) d^4 , low spin |

Choose the correct answer from the options given below:

- 1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- 2. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- 3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- 4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Which of the statements given below are correct with reference to corrosion:

- (A) Corrosion covers the surfaces of metallic objects with oxides or other salts.
- (B) In corrosion, a metal is oxidised by the loss of electrons to oxygen and results in the formation of oxides.
- (C) Rust is hydrated ferric oxide.
- (D) A sacrificial electrode is used to save an object from corrosion.

Choose the correct answer from the options given below.

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.5 The complex $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]\text{Cl}$ will show.....

- (A) Linkage isomerism
- (B) Ionization isomerism
- (C) Coordination isomerism
- (D) Geometrical isomerism

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Which of the following will not respond to Fehling's Test?

- (A) Benzaldehyde
- (B) Acetone
- (C) Acetaldehyde
- (D) Acetophenone

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (B) and (D) only
4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Enzymes are biological catalysts. They accelerate biological reactions. Which of the following is not true about enzymes?

1. They are globular proteins.
2. They are required in small quantities.
3. The names of all enzymes doesn't end with the suffix -ase.
4. Enzymes accelerate biological reactions by increasing the energy of activation.

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The calculated magnetic moment using the 'spin only' formula $\mu = \sqrt{n(n+2)}$ for Cr^{2+} and Fe^{2+} respectively is

1. 3.87, 4.89
2. Both 4.89
3. Both 3.87
4. 4.89, 3.87

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Which of the following cannot be explained by the open chain structure of glucose?

1. Glucose reacts with hydroxylamine
2. Glucose undergoes oxidation with nitric acid
3. Glucose reacts with acetic anhydride to form pentaacetyl glucose.
4. Glucose exists in two forms in nature: α and β .

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Which of the following compounds will give a pair of enantiomers upon nucleophilic substitution reaction with a hydroxide ion?

1. 2-bromobutane
2. 2-bromo-2-methylbutane
3. 1-bromo-3-methylbutane
4. 1-bromobutane

Options 1. 1

2. 2
3. 3
4. 4

Q.11 DNA and RNA do not differ from each other in which of the following

1. Nature of sugar
2. Nature of the nitrogenous bases
3. Nature of the backbone
4. Secondary structure

Options

1. 1
2. 2
3. 3
4. 4

Q.12 Identify the most volatile compound

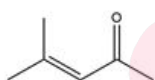
1. n-butanol
2. Triethylamine
3. Propylamine
4. Propane

Options

1. 1
2. 2
3. 3
4. 4

Q.13

Identify the correct IUPAC name of the given compound:



1. 4-Methylpent-3-en-2-one
2. 2-Methylpent-2-en-3-one
3. 1,1-dimethylbut-1-en-3-one
4. Mesityloxide

Options

1. 1
2. 2
3. 3
4. 4

Q.14 What is the product(s) formed when toluene reacts with Cl_2 in presence of anhydrous AlCl_3 ?

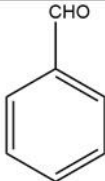
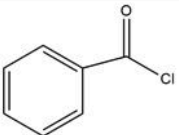
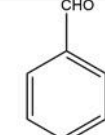
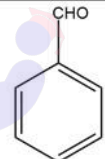
1. Benzyl Chloride
2. Benzal Chloride
3. ortho and para chlorotoluene
4. para Chlorotoluene

Options

1. 1
2. 2
3. 3
4. 4

Q.15

Match List-I with List-II

List-I	List-II
(Reaction Name)	(Reaction)
(A) Rosenmund Reaction	(I) $\text{C}_6\text{H}_5\text{CN} + \text{SnCl}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(I) HCl}} \text{C}_6\text{H}_5\text{CHO}$
(B) Stephen Reaction	(II)  $\xrightarrow[\text{Anhy. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$ 
(C) Gattermann Koch Reaction	(III)  $\xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2}$ 
(D) Etard Reaction	(IV)  $+ \text{CrO}_2\text{Cl}_2 \xrightarrow[\text{(II) H}_3\text{O}^+]{\text{(I) CS}_2}$ 

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.16 If you place blood cells in a solution containing more than 0.9% (mass/volume) sodium chloride, water will flow out of the cells and they will shrink. Such a solution is called

- Hypotonic
- Hypertonic
- Saturated
- Isotonic

Options 1. 1

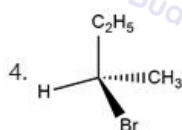
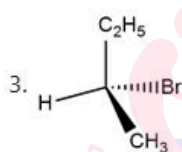
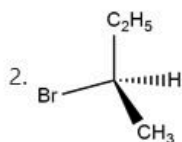
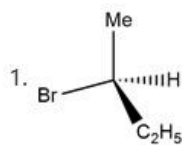
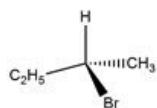
2. 2

3. 3

4. 4

Q.17

Identify the enantiomer of the molecule given below :



Options 1. 1

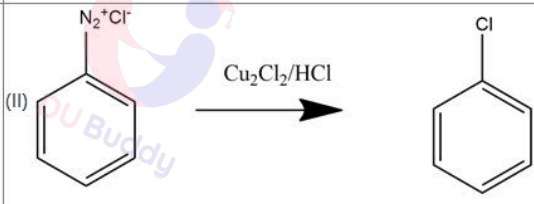
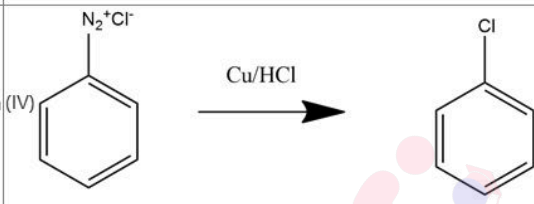
2. 2

3. 3

4. 4

Q.18

Match List-I with List-II

List-I	List-II
(Reaction Name)	(Reaction)
(A) Carbylamine reaction	(I) $\text{RNH}_2 + \text{KOH} + \text{CHCl}_3 \longrightarrow \text{RNC} + \text{KCl} + \text{H}_2\text{O}$
(B) Gatterman Reaction	(II) 
(C) Sandmeyer reaction	(III) $\text{RCONH}_2 + \text{NaOH} + \text{Br}_2 \longrightarrow \text{RNH}_2 + \text{NaBr} + \text{H}_2\text{O} + \text{Na}_2\text{CO}_3$
(D) Hoffmann Bromamide reaction	(IV) 

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 The unit for molar conductivity is

- $\text{S m}^2 \text{mol}^{-1}$
- S m^{-1}
- S m mol^{-1}
- $\text{S}^{-1} \text{m}^2 \text{mol}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Match List-I with List-II

List-I	List-II
(Name of the molecule/drug)	(Use/disease name)
(A) Thyroxine	(I) Scurvy
(B) Ascorbic acid	(II) Thyroid gland
(C) Steroid hormones	(III) Identification of criminals
(D) DNA fingerprinting	(IV) Adrenal cortex

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Which of the following statements is incorrect?

- chromate ion has tetrahedral geometry
- sodium and potassium dichromates are strong oxidising agents
- during oxidizing action in an acidic solution, $\text{Cr}_2\text{O}_7^{2-}$, gets converted to Cr^{2+}
- potassium dichromate is used as a primary standard in volumetric analysis.

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Which of the following alkyl halide will undergo S_N1 reaction fastest?

1. $\text{Me}_3\text{C-F}$
2. $\text{Me}_3\text{C-Cl}$
3. $\text{Me}_3\text{C-Br}$
4. $\text{Me}_3\text{C-I}$

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Which of the following statements is(are) correct for amino acids?

- (A) Amino acids behave like salts
- (B) In aqueous solution they form zwitter ion.
- (C) Most naturally occurring amino acids exist in D configuration
- (D) Tyrosine is a non-essential amino acid

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The reagent involved in the Clemmensen reduction of carbonyl compound is

1. Sodium amalgam and HCl
2. Zinc amalgam and HCl
3. KOH and ethylene glycol
4. Zinc amalgam and HNO_3

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The conductivity of electrolytic solutions does not depend upon

1. Nature of the electrolyte added
2. Nature of solvent and its viscosity
3. Pressure
4. Temperature

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Four different electrolyte solutions with the concentrations given below have the same molar conductivity value of $1.29 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$. Arrange them in increasing order of their conductivity values:

- (A) 0.1 M
- (B) 0.075 M
- (C) 0.15 M
- (D) 0.05 M

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (D), (B), (A), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Calculate the mole fraction of ethanoic acid (CH_3COOH) in an aqueous solution containing 20% of CH_3COOH by mass.

1. 0.0697
2. 0.3333

3. 4.444

4. 4.777

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Thiosulphate $S_2O_3^{2-}$ is oxidized to in presence of permanganate ion.

1. S^{2-}

2. SO_3^{2-}

3. SO_4^{2-}

4. SO_2^{2-}

Options 1. 1

2. 2

3. 3

4. 4

Q.29 In the process of extraction, gold combines with cyanide in the presence of oxygen and water to make a complex. The formula of the complex is

1. $[Au(CN)_2]^{2-}$

2. $[Au(CN)_2]^-$

3. $[Au(CN)_6]^{2-}$

4. $[Au(CN)_4]^{2-}$

Options 1. 1

2. 2

3. 3

4. 4

Q.30 Match the property given in List-I with the element given in List-II

List-I (Property)

List-II (Element)

(A) Lanthanoid, which can show +2 oxidation state

(I) Pm

(B) Radioactive lanthanoid

(II) Lu

(C) Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state

(III) Eu

(D) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state

(IV) Gd

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

4. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

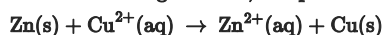
Options 1. 1

2. 2

3. 3

4. 4

Q.31 For the following reaction, at equilibrium the Nernst equation may be written as:



1. $E_{\text{Cell}}^{\circ} = \frac{2.303 RT}{F} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$

2. $E_{\text{Cell}}^{\circ} = \frac{2.303 RT}{2F} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$

3. $E_{\text{Cell}}^{\circ} = \frac{2.303 RT}{F} \log \frac{[Cu^{2+}]}{[Zn^{2+}]}$

4. $E_{\text{Cell}}^{\circ} = \frac{2.303 RT}{2F} \log \frac{[Cu^{2+}]}{[Zn^{2+}]}$

Options 1. 1

2. 2

3. 3

4. 4

Q.32 The boiling point elevation constant K_b is also known as:

1. Cryoscopic Constant

2. Molar Depression Constant

3. Ebullioscopic Constant

4. Boiling Constant

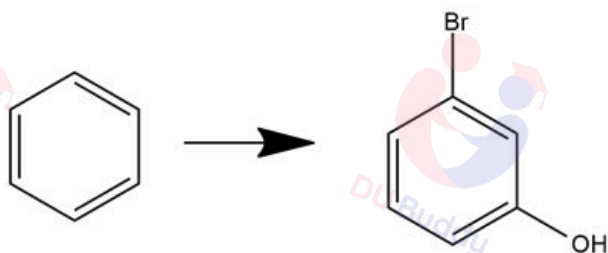
Options 1. 1

2. 2

3. 3

Q.33

Choose the correct sequence of reagents to carry out the following conversion:



- (A) Conc. HNO_3 /Conc. H_2SO_4
 (B) Sn/HCl
 (C) $\text{Br}_2/\text{FeBr}_3$
 (D) NaNO_2/HCl ; followed by boiling with water

Choose the **correct** answer from the options given below:

- (A), (C), (B), (D)
- (A), (B), (C), (D)
- (B), (A), (D), (C)
- (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 Arrange the following complexes in \textbf{increasing order} of their magnetic moment

- (A) $[\text{NiCl}_4]^{2-}$
 (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (C) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
 (D) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

Choose the correct answer from the options given below:

- (A), (B), (D), (C)
- (D), (A), (C), (B)
- (D), (C), (B), (A)
- (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Which compound will exhibit the lowest boiling point ?

- Diethylamine
- Butanol
- Butanamine
- N,N-dimethylethanamine

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Based on the data given below, which metal in its +2 oxidation state will not liberate hydrogen from acids ?

Metal	V	Ti	Co	Cr
$E^\circ (\text{M}^{3+}/\text{M}^{2+})$	-0.26	-0.37	+1.97	-0.41

- Co
- Cr
- V
- Ti

- Options 1. 1
2. 2
3. 3
4. 4

Q.37

Identify A and B in the following reaction.



1. A: Benzal Chloride; B: Benzaldehyde
2. A: Benzyl Chloride; B: Benzyl alcohol
3. A: *o*-Chlorotoluene; B: *o*-Cresol
4. A: *p*-Chlorotoluene; B: *p*-Cresol

- Options 1. 1
2. 2
3. 3
4. 4

Q.38 Which of the following statements regarding osmosis is not correct?

1. The direction of osmosis can be reversed if a pressure larger than the osmotic pressure is applied to the solution side.
2. Reverse osmosis is used in the desalination of sea water.
3. It involves the movement of solvent across semi-permeable membranes only.
4. It is common in liquids as well as in gases.

- Options 1. 1
2. 2
3. 3
4. 4

Q.39 Arrange the following in decreasing order of the basic character in aqueous phase

- (A) Ethanamine
- (B) Benzeneamine
- (C) Phenylmethanamine
- (D) *N*-methylaniline

Choose the correct answer from the Options given below:

1. (A), (C), (B), (D)
2. (A), (C), (D), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

- Options 1. 1
2. 2
3. 3
4. 4

Q.40 1.02 g of the following substances are dissolved in 100 mL of water. Arrange them in increasing order of their molarity:

- (A) NaCl
- (B) KCl
- (C) MgSO_4
- (D) NaOH

Choose the correct answer from the options given below:

1. (C), (B), (A), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (D), (A), (B), (C)

- Options 1. 1
2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the questions based on the passage:

The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide

simultaneously in order to bring about a chemical reaction is called molecularity of a reaction. In the rate equation

$$\text{Rate} = k[A]^x[B]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B, respectively. Sum of these exponents, i.e., $x + y$ gives the overall order of a reaction where x and y represent the order with respect to the reactants A and B respectively. Hence, the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. For a first order reaction, the concentration of the reactant varies as $[R] = [R]_0 e^{-kt}$

The units of first and zero order rate constant of reactions are respectively:

1. s^{-1}, Ms^{-1}
2. Ms^{-1}, s^{-1}
3. M, s^{-1}
4. s^{-1}, M

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the questions based on the passage:

The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction. In the rate equation

$$\text{Rate} = k[A]^x[B]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B, respectively. Sum of these exponents, i.e., $x + y$ gives the overall order of a reaction where x and y represent the order with respect to the reactants A and B respectively. Hence, the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. For a first order reaction, the concentration of the reactant varies as $[R] = [R]_0 e^{-kt}$

The conversion of molecules X to Y follows second order kinetics. If the concentration of X is increased to four times, how does it affect the rate of formation of Y?

1. The rate will decrease by 2 times
2. The rate will increase by 4 times
3. The rate will increase by 16 times
4. The rate will increase by 64 times

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the questions based on the passage:

The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction. In the rate equation

$$\text{Rate} = k[A]^x[B]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B, respectively. Sum of these exponents, i.e., $x + y$ gives the overall order of a reaction where x and y represent the order with respect to the reactants A and B respectively. Hence, the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. For a first order reaction, the concentration of the reactant varies as $[R] = [R]_0 e^{-kt}$

Which of the following is a zero order reaction?

1. Radioactive decay of unstable nuclei
2. Hydrogenation of ethene
3. Decomposition of gaseous ammonia on a hot platinum surface
4. Decomposition of N_2O_5

Options 1. 1

2. 2

3. 3

4. 4

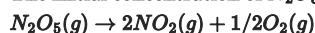
Q.44 Read the passage carefully and answer the questions based on the passage:

The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction. In the rate equation

$$\text{Rate} = k[A]^x[B]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B, respectively. Sum of these exponents, i.e., $x + y$ gives the overall order of a reaction where x and y represent the order with respect to the reactants A and B respectively. Hence, the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. For a first order reaction, the concentration of the reactant varies as $[R] = [R]_0 e^{-kt}$

The initial concentration of N_2O_5 in the following first order reaction



Extra close brace or missing open brace

1. 0.040 min^{-1}
2. 0.023 min^{-1}
3. 0.032 min^{-1}

4. 0.050 min^{-1}

- Options
- 1.
 - 2.
 - 3.
 - 4.

Q.45 Read the passage carefully and answer the questions based on the passage:

The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction. In the rate equation

$$\text{Rate} = k[A]^x[B]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B, respectively. Sum of these exponents, i.e., $x + y$ gives the overall order of a reaction where x and y represent the order with respect to the reactants A and B respectively. Hence, the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction. For a first order reaction, the concentration of the reactant varies as $[R] = [R]_0 e^{-kt}$

Which of the following statements is not true for order of a reaction?

1. A balanced chemical equation does not give the correct order of reaction every time.
2. Order of a reaction cannot be fraction.
3. Order of a reaction can be determined experimentally only.
4. Order of a reaction can be zero also.

- Options
- 1.
 - 2.
 - 3.
 - 4.

Q.46 Read the passage carefully and answer the questions based on the passage:

Alcohols are versatile compounds as they can act as both nucleophile and electrophile. They undergo reactions based on cleavage of O-H and C-O bonds. The cleavage of O-H bond reflects the acidic character of alcohols. However they are weaker acids than water, phenol and carboxylic acids. Alkoxide ion formed by loss of proton from the OH group of alcohols reacts with water to form the corresponding conjugate acid - base pair. They act as Bronsted base due to presence of unshared electron pairs on oxygen. Both alcohols and phenol react with carboxylic acids, acid chlorides and anhydrides to form esters. The introduction of acetyl group in alcohols or phenol is known as acetylation. Alcohols react with hydrogen halides to form alkyl halides which form the basis of Lucas test. They also undergo acid catalyzed dehydration to form alkenes. Alcohols can be easily oxidized with variety of oxidizing agents. The nature of the product depends on the type of oxidizing agent and the nature of alcohol.

Oxidation of alcohols involves formation of carbon oxygen double bond with

1. Cleavage of an O-H bond
2. Cleavage of C-H bond
3. Cleavage of O-H bond and C-H bond
4. Cleavage of C-C bond

- Options
- 1.
 - 2.
 - 3.
 - 4.

Q.47 Read the passage carefully and answer the questions based on the passage:

Alcohols are versatile compounds as they can act as both nucleophile and electrophile. They undergo reactions based on cleavage of O-H and C-O bonds. The cleavage of O-H bond reflects the acidic character of alcohols. However they are weaker acids than water, phenol and carboxylic acids. Alkoxide ion formed by loss of proton from the OH group of alcohols reacts with water to form the corresponding conjugate acid - base pair. They act as Bronsted base due to presence of unshared electron pairs on oxygen. Both alcohols and phenol react with carboxylic acids, acid chlorides and anhydrides to form esters. The introduction of acetyl group in alcohols or phenol is known as acetylation. Alcohols react with hydrogen halides to form alkyl halides which form the basis of Lucas test. They also undergo acid catalyzed dehydration to form alkenes. Alcohols can be easily oxidized with variety of oxidizing agents. The nature of the product depends on the type of oxidizing agent and the nature of alcohol.

Lucas reagent is

1. Conc. HCl and ZnCl_2
2. Conc. HCl
3. ZnCl_2
4. Dil. HCl and ZnCl_2

- Options
- 1.
 - 2.
 - 3.
 - 4.

Q.48 Read the passage carefully and answer the questions based on the passage:

Alcohols are versatile compounds as they can act as both nucleophile and electrophile. They undergo reactions based on cleavage of O-H and C-O bonds. The cleavage of O-H bond reflects the acidic character of alcohols. However they are weaker

acids than water, phenol and carboxylic acids. Alkoxide ion formed by loss of proton from the OH group of alcohols reacts with water to form the corresponding conjugate acid - base pair. They act as Bronsted base due to presence of unshared electron pairs on oxygen. Both alcohols and phenol react with carboxylic acids, acid chlorides and anhydrides to form esters. The introduction of acetyl group in alcohols or phenol is known as acetylation. Alcohols react with hydrogen halides to form alkyl halides which form the basis of Lucas test. They also undergo acid catalyzed dehydration to form alkenes. Alcohols can be easily oxidized with variety of oxidizing agents. The nature of the product depends on the type of oxidizing agent and the nature of alcohol.

Which of the following gives the correct sequential steps for dehydration of alcohol ?

1. Formation of carbocation, elimination of a proton, formation of protonated alcohol.
2. Formation of protonated alcohol, formation of carbocation, elimination of a proton
3. Elimination of proton, Formation of carbocation, protonation
4. Loss of water, formation of carbocation, elimination of proton

Options 1. 1
2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions based on the passage:

Alcohols are versatile compounds as they can act as both nucleophile and electrophile. They undergo reactions based on cleavage of O-H and C-O bonds. The cleavage of O-H bond reflects the acidic character of alcohols. However they are weaker acids than water, phenol and carboxylic acids. Alkoxide ion formed by loss of proton from the OH group of alcohols reacts with water to form the corresponding conjugate acid - base pair. They act as Bronsted base due to presence of unshared electron pairs on oxygen. Both alcohols and phenol react with carboxylic acids, acid chlorides and anhydrides to form esters. The introduction of acetyl group in alcohols or phenol is known as acetylation. Alcohols react with hydrogen halides to form alkyl halides which form the basis of Lucas test. They also undergo acid catalyzed dehydration to form alkenes. Alcohols can be easily oxidized with variety of oxidizing agents. The nature of the product depends on the type of oxidizing agent and the nature of alcohol.

Identify the most acidic alcohol among the following:

1. Butanol
2. Isobutanol
3. Tertiary butanol
4. Isopropanol

Options 1. 1
2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions based on the passage:

Alcohols are versatile compounds as they can act as both nucleophile and electrophile. They undergo reactions based on cleavage of O-H and C-O bonds. The cleavage of O-H bond reflects the acidic character of alcohols. However they are weaker acids than water, phenol and carboxylic acids. Alkoxide ion formed by loss of proton from the OH group of alcohols reacts with water to form the corresponding conjugate acid - base pair. They act as Bronsted base due to presence of unshared electron pairs on oxygen. Both alcohols and phenol react with carboxylic acids, acid chlorides and anhydrides to form esters. The introduction of acetyl group in alcohols or phenol is known as acetylation. Alcohols react with hydrogen halides to form alkyl halides which form the basis of Lucas test. They also undergo acid catalyzed dehydration to form alkenes. Alcohols can be easily oxidized with variety of oxidizing agents. The nature of the product depends on the type of oxidizing agent and the nature of alcohol.

Pyridinium chlorochromate is used for:

1. Dehydration of primary alcohol
2. Conversion of primary alcohol to aldehyde
3. Conversion of secondary alcohol to carboxylic acids
4. Conversion of Primary alcohols to carboxylic acids

Options 1. 1
2. 2
3. 3
4. 4